

Andrei Silviu Dragnea

Professional automator passionate about static analysis, refactoring and automating every aspect of the software development process.

A developer eager to make other developers' lives easier.

Education

2013-2017 **BSc, Computer Science and Engineering**; Faculty of Automatic Control and Computers, Politehnica University of Bucharest (Bucharest)

*Thesis title: Automatic Recursion Removal in Java (as an **IntelliJ Plugin**)*

Experience

- IntelliJ plugin **developer**
- Spring framework
- Async programming expert (Project Reactor from Spring, Kotlin coroutines, cats-effect from Scala, Rust async programming)
- Open source contributor to **Docker**, **mockito**, **intellij-kotlin**, **intellij-community**, **reactor-core**, **reactor-netty**, **netty**, **spring-framework**, Scala **kubernetes-client**
- Proficient in Java, Kotlin, Scala, Rust, C and Python
- Build system expert: Maven, Gradle, Earthly, Docker, sbt
- Author of an automatic Git **rebaser** written in Rust
- Passionate about static analysis and automatic refactoring in Java, Kotlin, Scala and Rust
- Working on a **converter** from Java reactive code to Kotlin coroutines
- Experience with enterprise identity protocols and frameworks, such as OAuth2, SCIM2, SAML
- Worked with Docker, Kubernetes, Envoy, Grafana, Prometheus, AWS EC2, S3, Route53, Cloudwatch, Splunk, Datadog, NewRelic, Azure Active Directory
- Good knowledge of Operating Systems internals and how async frameworks are implemented
- Promoter of the **clean tests** paradigm

May 2021 - Present **Software Engineer at Adobe (Media Analytics team)** I work on a performance-sensitive system collecting Analytics events from video players in web browsers. I optimized the runtime performance by correctly handling blocking calls, thus avoiding starving the **cats-effect** compute pool. I also enforced strong static analysis checks on the Scala project at compile time, in order to maintain the high quality of the code. I re-wrote the event ingestion system in Rust as a POC, resulting into 2.6x CPU usage and 50x memory usage reduction.

March 2018 - May 2021 **Software Engineer at Adobe (Identity Management Services team)** I was part of the Identity Management Services team. I worked on implementing the server

side of the SCIM2 protocol over a Spring 5 reactive stack. I developed a thin wrapper over Mockito and Spring TestContext framework in order to promote writing **clean tests**. I introduced Kotlin and coroutines to our team, as a developer-friendly alternative to writing non-blocking code. I made major software architectural changes by using smart refactoring tricks, resulting in a simpler system design.

**September
2017 - De-
cember 2017**

Software Engineer at Instacar I implemented a TCP server over a proprietary binary protocol for remote car control. I recompiled the Linux kernel with custom TCP settings in order to compensate for a faulty TCP implementation on the devices used for remote car control. I introduced type hints to the Python 3 codebase and I developed a Pycharm **plugin** for dead code elimination.

I always strive to find the simplest solution possible to a problem. I proactively fix problems from any software project that I encounter, whether it is internal to my company, or an open source project.

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